

Generation Interconnection System Impact Study Report

For Queue Project Q17-0902

Plumosus 138 kV

May 16, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0902. The Interconnection Customer proposes a 284.1 MW solar photovoltaic facility interconnecting at the Plumosus 138 kV substation in the Pennsylvania area. The request was received on December 13, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 332 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 15 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$135,765,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Powhatan - Vinoy 138 kV L	Plumosus - Cox Battery Energy Storage 138 kV L	166.1	167	100.0
J J Elmore - Powhatan 138 kV Line	Cox Battery Energy Storage - Plumosus 138 kV	164.1	166	100.0
Axford - Empire 138 kV	Southbase - Milwaukee 138 kV	161.6	273	6.3
Vinoy - Plumosus 138 kV	Plumosus - Cox Battery Energy Storage 138 kV L	158.4	178	100.0
Asheboro Substation - Empire 138 kV	Southbase - Milwaukee 138 kV	157.3	283	6.3
Dos Marinas 2003 - Brent 138 kV	Southbase - Kings 138 kV	153.2	275	5.6
Rewey Substation 345/138kV Transformer Bank	Orchard Rd - Ft Humbug 500 kV	144.3	446	12.3
Southbase - Kings 138 kV L	Dos Marinas 2003 - Brent 138 kV	144.2	331	5.5
Plumosus - Cox Battery Energy Storage 138 kV L	Vinoy - Plumosus 138 kV	141.2	200	100.0
Axford - Brent 138 kV	Milwaukee - Southbase Substation 138 kV	140.7	312	6.3
Rewey - Loco Hills 138 kV Line	Milwaukee - Rewey 138 kV L	140.2	180	6.1
Milwaukee - Rewey 138 kV	Asheboro - Empire 138 kV	139.1	307	7.2
J J Elmore - Searsburg 138 kV	Cox Battery Energy Storage Substation - Plumosus 138 kV	137.5	197	100.0
Ft Humbug 345 kV/138 kV XFMR	W Campus - Ft Humbug 345 kV	135.6	356	5.0
Omro - Clark Rd 138 kV	William F Matso Generating Sta 138-1 kV Autotransformer	104.9	225	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Vinoy - Powhatan 138 kV	Rebuild line with higher-capacity conductor	\$3,102,000	100% (\$3,102,000)
J J Elmore - Powhatan 138 kV Line	Rebuild line with higher-capacity conductor	\$10,284,000	100% (\$10,284,000)
Axford - Empire 138 kV	Rebuild line with higher-capacity conductor	\$3,029,000	100% (\$3,029,000)
Vinoy - Plumosus 138 kV	Rebuild line with higher-capacity conductor	\$2,180,000	100% (\$2,180,000)
Empire - Asheboro 138 kV	Rebuild line with higher-capacity conductor	\$13,465,000	100% (\$13,465,000)
Brent - Dos Marinas 2003 138 kV L	Rebuild line with higher-capacity conductor	\$27,981,000	100% (\$27,981,000)
Rewey 345/138kV TX	Replace transformer with higher-rated unit	\$13,019,000	81% (\$10,592,000)
Kings - Southbase Substation 138 kV	Rebuild line with higher-capacity conductor	\$9,609,000	78% (\$7,478,000)
Cox Battery Energy Strage - Plumosus 138 kV	Rebuild line with higher-capacity conductor	\$16,686,000	100% (\$16,686,000)
Axford - Brent 138 kV line	Rebuild line with higher-capacity conductor	\$3,917,000	100% (\$3,917,000)
Rewey - Loco Hills 138 kV Line	Rebuild line with higher-capacity conductor	\$19,340,000	100% (\$19,340,000)
Milwaukee - Rewey 138 kV line	Rebuild line with higher-capacity conductor	\$3,835,000	68% (\$2,615,000)
J J Elmore - Searsburg 138 kV	Rebuild line with higher-capacity conductor	\$5,132,000	100% (\$5,132,000)
Ft Humbug 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$11,504,000	56% (\$6,487,000)
Omro - Clark Rd 138 kV	Reconductor line	\$3,477,000	100% (\$3,477,000)
	Total allocated to project		\$135,765,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.